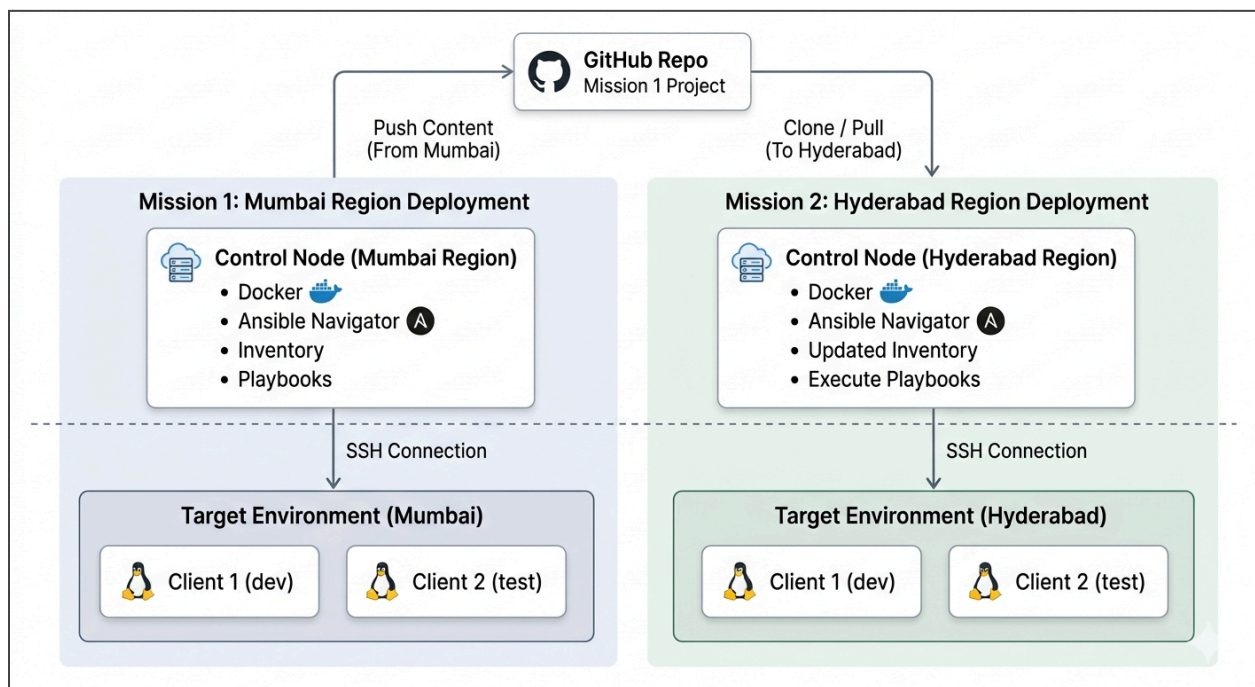


# DevOps Practical Mission

## Objective

The main objective of this project is to gain practical experience in AWS infrastructure deployment and infrastructure automation using Ansible. The project aims to automate software installation, web server configuration, content management, and server administration while using Git for version control. It also demonstrates how an existing automation project can be reused to recreate an identical environment in another AWS region with minimal manual effort.

## Architecture Diagram



## Prerequisites

- AWS EC2 Instance
- Amazon Linux 2023
- Docker
- Ansible Navigator
- Git & GitHub
- SSH

# Workflow Explanation

## MISSION 1

1. Launch three **AWS EC2** instances in the **Mumbai Region**.
2. Create the **devops** user and configure **SSH** key-based authentication.
3. Install **Docker** and **Ansible Navigator** on the Control Node.
4. Configure the Ansible project using the **inventory** and **ansible.cfg** files.
5. Create **Ansible playbooks**, **roles**, and **Jinja2** templates to automate server configuration.
6. Execute all playbooks using **Ansible Navigator** inside a **Docker container**.
7. Verify package installation, Apache configuration, and web server output.
8. **Push** the automation project to the **GitHub** repository.

## MISSION 2

9. Create a new AWS environment in the **Hyderabad Region**.
10. **Clone** the **GitHub** repository and update the inventory.
11. Re-execute all playbooks to recreate the infrastructure.
12. Verify the successful deployment on the new environment.

## **MISSION 1: Build and Configure an Ansible Automation Environment**

### **Phase 1: Infrastructure Setup**

#### **Task 1: Create AWS Infrastructure**

##### **Requirements**

1. Created three AWS EC2 instances in the Mumbai Region: one Control Node (**mumbai-control**) and two Client Nodes (**mumbai-client1** and **mumbai-client2**).
2. Create a user named devops on all machines.
3. Configure SSH key-based authentication for the devops user.
4. Use the following project directory on the control node: /home/devops/ansible
5. All Ansible playbooks must be executed using:
  - Docker
  - ansible-navigator

#### **Step 01: Created three AWS EC2 instances in Mumbai Region**

Control Node: **mumbai-control** | Client Nodes: **mumbai-client 1**, **mumbai-client 2**



## Step 2: Configure SSH Key-Based Authentication

Configured SSH key-based authentication for the **devops** user to enable passwordless communication between the Control Node and Client Nodes.

### In Control Node

- Generated an SSH key pair on the Control Node.
- Copied the public key to both Client Nodes.

```
[ec2-user@ip-172-31-43-214 ~]$ sudo su - devops
[devops@ip-172-31-43-214 ~]$ ssh-keygen
Generating public/private rsa key pair.
Enter file in which to save the key (/home/devops/.ssh/id_rsa):
Created directory '/home/devops/.ssh'.
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/devops/.ssh/id_rsa
Your public key has been saved in /home/devops/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:vGdZwssKtQs36vq/N/gzTnqv4N2L3L2x5BP2akdv2WI devops@ip-172-31-43-214.ap-south-1.compute.internal
The key's randomart image is:
+---[RSA 3072]-----+
|
|  .
| o . .
| o S . + +
| = * = *Eo+
| . =oB =.+++
| . .oBB. +.o+
| .ooo=*Bo..o
+---[SHA256]-----+
[devops@ip-172-31-43-214 ~]$ cat ~/.ssh/id_rsa.pub
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQOC0ex22MqvlztVmI81+Tue4gcYWCpQOxb6o4mb70xxvCrRfnaApeU7rORDeXYEEID31QO7WShTSc83LMSR3VwY2M/1NnitaFp
60692Cj48R2XANobYrdsOag6ni3yzi3HIdBK/T0BI0CSLHwY48Isv6negtXGwEXRK/CT6m/dEqc5NVqdIv61KjdzzFaxNO6wC641rn5WY5IveldewgbbBJulJ4mU8HCSEf9iDM
JalfP20V5jEMr55vlt1VnMtpTc+IjBfmjiY4KLxyzcmFwer5K/8OVJsoPNm3GoGMbwn39kly9NdD4MFB7tHKqgH1PIOD9InSOMpSLziWlKctdzhylrzzRpc7A36bhiJ0qHUU
G9SNnS45TWpuxKtziHGMdp90UF+3gxdkYsmh6qfRp2nOXlob2PRYG8C7FfBRrYUmiO+y9NpF5boZFHzNYG2hts6Usiml0py83O6bbn09WFEjZB2VYYXm+6TNxtFW2WTjUv3nZ
JMVacmsAgHpKBpc= devops@ip-172-31-43-214.ap-south-1.compute.internal
[devops@ip-172-31-43-214 ~]$
```

### In Client Node - 1 & 2

- Creating `authorized_keys` and setting permissions

```
[ec2-user@ip-172-31-42-176 ~]$ sudo mkdir -p /home/devops/.ssh
[ec2-user@ip-172-31-42-176 ~]$ sudo vi /home/devops/.ssh/authorized_keys
[ec2-user@ip-172-31-42-176 ~]$ sudo chown -R devops:devops /home/devops/.ssh
[ec2-user@ip-172-31-42-176 ~]$ sudo chmod 700 /home/devops/.ssh
[ec2-user@ip-172-31-42-176 ~]$ sudo chmod 600 /home/devops/.ssh/authorized_keys
[ec2-user@ip-172-31-42-176 ~]$
```

```
[ec2-user@ip-172-31-46-37 ~]$ sudo mkdir -p /home/devops/.ssh
[ec2-user@ip-172-31-46-37 ~]$ sudo vi /home/devops/.ssh/authorized_keys
[ec2-user@ip-172-31-46-37 ~]$ sudo chown -R devops:devops /home/devops/.ssh
[ec2-user@ip-172-31-46-37 ~]$ sudo chmod 700 /home/devops/.ssh
[ec2-user@ip-172-31-46-37 ~]$ sudo chmod 600 /home/devops/.ssh/authorized_keys
[ec2-user@ip-172-31-46-37 ~]$
```

Verify passwordless SSH connectivity from the Control Node to the Client Node using SSH.

From Control Node - # ssh devops@<public ip\_Client 1>

```
[devops@ip-172-31-43-214 ~]$ ssh devops@13.206.238.172
The authenticity of host '13.206.238.172 (13.206.238.172)' can't be established.
ED25519 key fingerprint is SHA256:ItylmQ4yKP6WwEoHXt9hx/6Hed2tnfxQ7MhHBCRS/4M.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '13.206.238.172' (ED25519) to the list of known hosts.

#_
~\#### Amazon Linux 2023
~~\#####\
~~\####|
~~\#/ https://aws.amazon.com/linux/amazon-linux-2023
~~V~'-'>
~~~
~~~.
~~~/_/
~~~/_m/'
```

→ Successful SSH login to Client Node 1 

From Control Node - # ssh devops@<public ip\_Client 2>

```
[devops@ip-172-31-43-214 ~]$ ssh devops@13.126.99.147
The authenticity of host '13.126.99.147 (13.126.99.147)' can't be established.
ED25519 key fingerprint is SHA256:ikHJlo+tTnUAazZtqcdVwsKeXxiz8cqJd6SHgGrcHi+k.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '13.126.99.147' (ED25519) to the list of known hosts.
```

```
#_
~\##### Amazon Linux 2023
~~\_#####\
~~\_####|
~~\_#/
~~V~'|'-> https://aws.amazon.com/linux/amazon-linux-2023
~~~~
~~~.-.-./-/-/
~/m/'-/-/
```

```
[devops@ip-172-31-46-37 ~]$
```

→ Successful SSH login to Client Node 2 ✓

Verified SSH connectivity successfully.

### Step 3: Create the Ansible Project Directory

```
[devops@ip-172-31-43-214 ~]$ mkdir -p /home/devops/ansible
[devops@ip-172-31-43-214 ~]$ cd /home/devops/ansible
[devops@ip-172-31-43-214 ansible]$ pwd
/home/devops/ansible
[devops@ip-172-31-43-214 ansible]$
```

## Step 4: Set Up the Ansible Execution Environment

1. Installed Docker on the Control Node. `# Sudo dnf install docker -y`

```
devops@ip-172-31-43-214:~/ansible
[devops@ip-172-31-43-214 ansible]$ sudo dnf install docker -y
Amazon Linux 2023 Kernel Livepatch repository                                481 kB/s | 55 kB    00:00
Last metadata expiration check: 0:00:01 ago on Wed Jun 24 09:07:22 2026.
Dependencies resolved.
=====
Package                        Architecture      Version           Repository        Size
=====
Installing:
docker                        x86_64            25.0.16-1.amzn2023.0.2  amazonlinux      46 M
Installing dependencies:
container-selinux            noarch            4:2.245.0-1.amzn2023  amazonlinux       58 k
containerd                   x86_64            2.2.4-1.amzn2023.0.3  amazonlinux      24 M
iptables-libs                x86_64            1.8.8-3.amzn2023.0.2  amazonlinux      401 k
iptables-nft                 x86_64            1.8.8-3.amzn2023.0.2  amazonlinux      183 k
libcgroup                    x86_64            3.0-1.amzn2023.0.1    amazonlinux       75 k
libnetfilter_conntrack       x86_64            1.0.8-2.amzn2023.0.2  amazonlinux       58 k
libnftnl                     x86_64            1.0.1-19.amzn2023.0.2 amazonlinux       30 k
libnftnl                     x86_64            1.2.2-2.amzn2023.0.2  amazonlinux       84 k
pigz                         x86_64            2.5-1.amzn2023.0.4    amazonlinux       83 k
runc                         x86_64            1.3.5-1.amzn2023.0.2  amazonlinux      3.9 M
=====
Transaction Summary
=====
Install 11 Packages

Total download size: 76 M
Installed size: 284 M
Downloading Packages:
(1/11): container-selinux-2.245.0-1.amzn2023.noarch.rpm                1.5 MB/s | 58 kB    00:00
(2/11): iptables-libs-1.8.8-3.amzn2023.0.2.x86_64.rpm                8.9 MB/s | 401 kB  00:00
(3/11): iptables-nft-1.8.8-3.amzn2023.0.2.x86_64.rpm                 4.4 MB/s | 183 kB  00:00
(4/11): libcgroup-3.0-1.amzn2023.0.1.x86_64.rpm                     1.8 MB/s | 75 kB   00:00
(5/11): libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64.rpm       1.7 MB/s | 58 kB   00:00
(6/11): libnftnl-1.0.1-19.amzn2023.0.2.x86_64.rpm                   799 kB/s | 30 kB   00:00
(7/11): libnftnl-1.2.2-2.amzn2023.0.2.x86_64.rpm                     2.2 MB/s | 84 kB   00:00
(8/11): containerd-2.2.4-1.amzn2023.0.3.x86_64.rpm                   54 MB/s | 24 MB    00:00
(9/11): pigz-2.5-1.amzn2023.0.4.x86_64.rpm                           469 kB/s | 83 kB   00:00
(10/11): runc-1.3.5-1.amzn2023.0.2.x86_64.rpm                       63 MB/s | 3.9 MB   00:00
```

Verified the Docker installation

```
[devops@ip-172-31-43-214 ansible]$ sudo systemctl enable docker
Created symlink /etc/systemd/system/multi-user.target.wants/docker.service → /usr/lib/systemd/system/docker.service.
[devops@ip-172-31-43-214 ansible]$ sudo systemctl start docker
[devops@ip-172-31-43-214 ansible]$ docker --version
Docker version 25.0.14, build 0bab007
[devops@ip-172-31-43-214 ansible]$
```

2. **Ansible Navigator** installed on Control Node (Install Python pip and Install Ansible Navigator) 

`# sudo dnf install python3-pip -y`

- Installed **Python pip**, which is required to install Ansible Navigator. .

`# pip3 install ansible-navigator`

- Installed **Ansible Navigator** using pip3 to execute Ansible playbooks in the required container-based environment.

3. Verified the **Ansible Navigator** installation.

```
[devops@ip-172-31-43-214 ansible]$ ansible-navigator --version
ansible-navigator 24.2.0
[devops@ip-172-31-43-214 ansible]$
```

## Task 2: Install and Configure Ansible Environment

### Requirements

1. Create the inventory file in the Ansible project directory.
  - Add mumbai-client1 to the dev group.
  - Add mumbai-client2 to the test group.
2. Create the ansible.cfg configuration file.
3. Verify connectivity between the Control Node and Client Nodes using Ansible.

### Step 1: Create the Inventory File

Created the inventory file and added the client nodes to the dev and test groups with the required SSH user.

```
[devops@ip-172-31-43-214 ansible]$ vi inventory
[devops@ip-172-31-43-214 ansible]$ cat inventory
[dev]
client1 ansible_host=172.31.42.176

[test]
client2 ansible_host=172.31.46.37

[all:vars]
ansible_user=devops
[devops@ip-172-31-43-214 ansible]$
```

### Step 2: Configure the Ansible Configuration File

Configured the ansible.cfg file by specifying the inventory file and disabling host key checking.

```
[devops@ip-172-31-43-214 ansible]$ vi ansible.cfg
[devops@ip-172-31-43-214 ansible]$ cat ansible.cfg
[defaults]
inventory=inventory
host_key_checking=False
[devops@ip-172-31-43-214 ansible]$
```

### Step 3: Create the Ansible Test Playbook

Created a simple ping playbook to verify connectivity between the Control Node and the Client Nodes.



```
[devops@ip-172-31-43-214 ansible]$ vi ping.yml
[devops@ip-172-31-43-214 ansible]$ cat ping.yml
---
- hosts: all
  tasks:
    - ping:
[devops@ip-172-31-43-214 ansible]$
```

## Step 4: Configure Docker Permissions

Added the devops user to the docker group to allow Docker commands without requiring root privileges.

```
[devops@ip-172-31-43-214 ansible]$ sudo usermod -aG docker devops
[devops@ip-172-31-43-214 ansible]$ exit
logout
[ec2-user@ip-172-31-43-214 ~]$ sudo su - devops
Last login: Wed Jun 24 09:06:20 UTC 2026 on pts/1
[devops@ip-172-31-43-214 ~]$ groups
devops wheel docker
[devops@ip-172-31-43-214 ~]$ docker ps
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS          NAMES
[devops@ip-172-31-43-214 ~]$ cd /home/devops/ansible
```

## Step 5: Verify Ansible Connectivity

Executed the ping playbook using Ansible Navigator and verified successful communication with all managed nodes.

```
devops@ip-172-31-43-214:~/ansible
[devops@ip-172-31-43-214 ansible]$ ansible-navigator run ping.yml -m stdout
-----
Execution environment image and pull policy overview
-----
Execution environment image name:   ghcr.io/ansible/creator-ee:v0.22.0
Execution environment image tag:    v0.22.0
Execution environment pull arguments: None
Execution environment pull policy:  tag
Execution environment pull needed:  True
-----
Updating the execution environment
-----
Running the command: docker pull ghcr.io/ansible/creator-ee:v0.22.0
v0.22.0: Pulling from ansible/creator-ee
ca6ff2c3ecc4: Pull complete
2941e6f1d2bc: Pull complete
32db66e6bbce: Pull complete
92e1f4a92098: Pull complete
600b306c9749: Pull complete
2764fac24a7d: Pull complete
d419ad167630: Pull complete
7b03b9e3c798: Pull complete
05a232cb5cea: Pull complete
be915b595e0a: Pull complete
25380a2f785a: Pull complete
1a57afb20498: Pull complete
09f2372dbd92: Pull complete
Digest: sha256:2edd3559b9fa2f3a386d078bff531874a1d30455f8171d027cf3a9c7f50118d4
Status: Downloaded newer image for ghcr.io/ansible/creator-ee:v0.22.0
ghcr.io/ansible/creator-ee:v0.22.0

PLAY [all] *****

TASK [Gathering Facts] *****
[WARNING]: Platform linux on host client2 is using the discovered Python
interpreter at /usr/bin/python3.9, but future installation of another Python
interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-
```



```

core/2.16/reference_appendices/interpreter_discovery.html for more information.
ok: [client2]
[WARNING]: Platform linux on host client1 is using the discovered Python
interpreter at /usr/bin/python3.9, but future installation of another Python
interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-
core/2.16/reference_appendices/interpreter_discovery.html for more information.
ok: [client1]

TASK [ping] *****
ok: [client1]
ok: [client2]

PLAY RECAP *****
client1      : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
client2      : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
[devops@ip-172-31-43-214 ansible]$

```

## Task 3: Package Management Automation

### Requirements

1. Create the `packages.yml` playbook in the Ansible project directory.
2. Install the `mariadb` and `php` packages on the `dev` and `test` groups.
3. Install the **Development Tools** package group on the `dev` group.
4. Update all installed packages on the `dev` group.
5. Verify successful playbook execution and package installation.

### Step 1: Create and Configure the Package Management Playbook

Create the `packages.yml` playbook in the Ansible project directory and configure it to:

```

[devops@ip-172-31-43-214 ansible]$ vi packages.yml
[devops@ip-172-31-43-214 ansible]$ cat packages.yml
---
- hosts: dev
  become: yes

  tasks:

    - name: Update all packages
      dnf:
        name: "*"
        state: latest

    - name: Install mariadb
      dnf:
        name: mariadb
        state: present

    - name: Install php
      dnf:
        name: php
        state: present

    - name: Install Development Tools
      dnf:
        name: "@Development Tools"
        state: present

- hosts: test
  become: yes

  tasks:

    - name: Install mariadb
      dnf:
        name: mariadb
        state: present

```

```
- name: Install php
  dnf:
    name: php
    state: present
[devops@ip-172-31-43-214 ansible]$
```

- Install **mariadb** and **php** on the dev and test groups.
- Install the Development Tools package group on the dev group.
- Update all packages on the dev group.

## Step 2: Execute and Verify the Playbook

Run the playbook using Ansible Navigator to install and update the required packages on the target nodes. Verify successful execution by checking the task results and the final PLAY RECAP, ensuring all hosts completed successfully without failures.

```
[devops@ip-172-31-43-214 ansible]$ ansible-navigator run packages.yml -m stdout
PLAY [dev] *****
TASK [Gathering Facts] *****
[WARNING]: Platform linux on host client1 is using the discovered Python
interpreter at /usr/bin/python3.9, but future installation of another Python
interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-
core/2.16/reference_appendices/interpreter_discovery.html for more information.
ok: [client1]
TASK [Update all packages] *****
ok: [client1]
TASK [Install mariadb] *****
ok: [client1]
TASK [Install php] *****
ok: [client1]
TASK [Install Development Tools] *****
ok: [client1]
PLAY [test] *****
TASK [Gathering Facts] *****
[WARNING]: Platform linux on host client2 is using the discovered Python
interpreter at /usr/bin/python3.9, but future installation of another Python
interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-
core/2.16/reference_appendices/interpreter_discovery.html for more information.
ok: [client2]
TASK [Install mariadb] *****
changed: [client2]
```

```
TASK [Install php] *****
changed: [client2]
PLAY RECAP *****
client1      : ok=5    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
client2      : ok=3    changed=2    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
[devops@ip-172-31-43-214 ansible]$
```

## Task 4: Create and Use Roles

### Requirements

1. Create the Ansible role **myrole** in the project directory.
2. Configure the role to install and configure the Apache Web Server.
3. Create the **index.j2** template to display the hostname and IP address.

4. Create the `myrole.yml` playbook to execute the role.
5. Configure the playbook to run only on the `test` group.
6. Verify the web server deployment by accessing the web page using a browser or `curl`.

## Step 1: Configure the Ansible Role

Configure the `myrole` role by creating the `tasks/main.yml` file to install and start the Apache web server, then deploy the web page using a Jinja2 template.

```
[devops@ip-172-31-43-214 ansible]$ vi roles/myrole/tasks/main.yml
[devops@ip-172-31-43-214 ansible]$ cat roles/myrole/tasks/main.yml
---
- name: Install Apache
  dnf:
    name: httpd
    state: present

- name: Start Apache
  service:
    name: httpd
    state: started
    enabled: yes

- name: Deploy index page
  template:
    src: index.j2
    dest: /var/www/html/index.html
[devops@ip-172-31-43-214 ansible]$
```

## Step 2: Create the Jinja2 Template

Create the `index.j2` template to display the hostname and IP address on the web page.

```
[devops@ip-172-31-43-214 ansible]$ vi roles/myrole/templates/index.j2
[devops@ip-172-31-43-214 ansible]$ cat roles/myrole/templates/index.j2
<h1>Welcome to {{ inventory_hostname }}</h1>
<h2>Managed by Ansible Role</h2>
[devops@ip-172-31-43-214 ansible]$
```

## Step 3: Create and Execute the Playbook

Create the `myrole.yml` playbook to execute the `myrole` role on the `test` group, then run the playbook using Ansible Navigator.

```
[devops@ip-172-31-43-214 ansible]$ vi myrole.yml
[devops@ip-172-31-43-214 ansible]$ cat myrole.yml
---
- hosts: test
  become: yes

  roles:
    - myrole
[devops@ip-172-31-43-214 ansible]$
```

```
[devops@ip-172-31-43-214 ansible]$ ansible-navigator run myrole.yml -m stdout

PLAY [test] *****

TASK [Gathering Facts] *****
[WARNING]: Platform linux on host client2 is using the discovered Python
interpreter at /usr/bin/python3.9, but future installation of another Python
interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-
core/2.16/reference_appendices/interpreter_discovery.html for more information.
ok: [client2]

TASK [myrole : Install Apache] *****
ok: [client2]

TASK [myrole : Start Apache] *****
changed: [client2]

TASK [myrole : Deploy index page] *****
changed: [client2]

PLAY RECAP *****
client2      : ok=4    changed=2    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
[devops@ip-172-31-43-214 ansible]$
```

## Step 4: Verify the Deployment

Verify that the Apache web server is running and confirm the web page is accessible using the `curl` command.

```
# curl http://localhost
```

```
[ec2-user@ip-172-31-46-37 ~]$ curl http://localhost
<h1>Welcome to client2</h1>
<h2>Managed by Ansible Role</h2>
[ec2-user@ip-172-31-46-37 ~]$
```

## Task 5: Content Modification

### Requirements

1. Create the `issue.yml` playbook in the Ansible project directory.
2. Modify the contents of the `/etc/issue` file.
3. Set the content to **"development"** for the **dev** group.
4. Set the content to **"test"** for the **test** group.
5. Verify the updated `/etc/issue` content on all target nodes.

## Step 1: Create the Playbook

Create the `issue.yml` playbook and configure it to update the `/etc/issue` file with **"development"** for the `dev` group and **"test"** for the `test` group.

```
[devops@ip-172-31-43-214 ansible]$ vi issue.yml
[devops@ip-172-31-43-214 ansible]$ cat issue.yml
---
- hosts: dev
  become: yes

  tasks:
    - name: Set issue file for dev group
      copy:
        content: "development\n"
        dest: /etc/issue

- hosts: test
  become: yes

  tasks:
    - name: Set issue file for test group
      copy:
        content: "test\n"
        dest: /etc/issue
[devops@ip-172-31-43-214 ansible]$
```

## Step 2: Execute the Playbook

Run the `issue.yml` playbook using Ansible Navigator to apply the changes on the target nodes and verify that the playbook executes successfully.

```
[devops@ip-172-31-43-214 ansible]$ ansible-navigator run issue.yml -m stdout
PLAY [dev] *****
TASK [Gathering Facts] *****
[WARNING]: Platform linux on host client1 is using the discovered Python
interpreter at /usr/bin/python3.9, but future installation of another Python
interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-
core/2.16/reference_appendices/interpreter_discovery.html for more information.
ok: [client1]

TASK [Set issue file for dev group] *****
changed: [client1]

PLAY [test] *****
TASK [Gathering Facts] *****
[WARNING]: Platform linux on host client2 is using the discovered Python
interpreter at /usr/bin/python3.9, but future installation of another Python
interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-
core/2.16/reference_appendices/interpreter_discovery.html for more information.
ok: [client2]

TASK [Set issue file for test group] *****
changed: [client2]

PLAY RECAP *****
client1      : ok=2    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
client2      : ok=2    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
[devops@ip-172-31-43-214 ansible]$
```

## Step 3: Verify the Configuration

Verify the updated `/etc/issue` content on both the **dev** and **test** nodes using the `cat /etc/issue` command.

**Client 1 (dev)** → (`cat /etc/issue` → `development`)

```
[ec2-user@ip-172-31-42-176 ~]$ cat /etc/issue
development
[ec2-user@ip-172-31-42-176 ~]$
```

**Client 2 (test)** → (`cat /etc/issue` → `test`)

```
[ec2-user@ip-172-31-46-37 ~]$ cat /etc/issue
test
[ec2-user@ip-172-31-46-37 ~]$
```

## Task 6: Custom-Based Web Server

### Requirements

1. Create the `custom.yml` playbook in the Ansible project directory.
2. Configure the Apache Web Server on the **dev** group.
3. Set the document root to `/webdev`.
4. Create a symbolic link from `/webdev` to `/var/www/html`.
5. Configure the web page to display the content **"development"**.
6. Verify the web page and confirm the displayed content.

### Step 1: Create the Playbook

Create the `custom.yml` playbook to install and configure the Apache Web Server, create the `/webdev` document root, add the web page content, and create a symbolic link from `/webdev` to `/var/www/html`.

```
[devops@ip-172-31-43-214 ansible]$ vi custom.yml
[devops@ip-172-31-43-214 ansible]$ cat custom.yml
---
- hosts: dev
  become: yes

  tasks:

    - name: Install Apache
      dnf:
        name: httpd
        state: present

    - name: Start Apache
      service:
        name: httpd
        state: started
        enabled: yes

    - name: Create webdev directory
      file:
        path: /webdev
        state: directory

    - name: Create index page
      copy:
        content: "development\n"
        dest: /webdev/index.html

    - name: Remove default html directory
      file:
        path: /var/www/html
        state: absent
```

```
    - name: Create symbolic link
      file:
        src: /webdev
        dest: /var/www/html
        state: link
[devops@ip-172-31-43-214 ansible]$ █
```

## Step 2: Execute the Playbook

Run the `custom.yml` playbook using Ansible Navigator and verify that all tasks complete successfully without errors.

```
[devops@ip-172-31-43-214 ansible]$ ansible-navigator run custom.yml -m stdout

PLAY [dev] *****

TASK [Gathering Facts] *****
[WARNING]: Platform linux on host client1 is using the discovered Python
interpreter at /usr/bin/python3.9, but future installation of another Python
interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-
core/2.16/reference_appendices/interpreter_discovery.html for more information.
ok: [client1]

TASK [Install Apache] *****
ok: [client1]

TASK [Start Apache] *****
changed: [client1]

TASK [Create webdev directory] *****
changed: [client1]

TASK [Create index page] *****
changed: [client1]

TASK [Remove default html directory] *****
changed: [client1]

TASK [Create symbolic link] *****
changed: [client1]

PLAY RECAP *****
client1 : ok=7 changed=5 unreachable=0 failed=0 skipped=0 rescued=0 ignored=0
[devops@ip-172-31-43-214 ansible]$ █
```



## Step 3: Verify the Web Server Configuration

Verify that the symbolic link is created, the web page content is available, and the Apache Web Server is serving the page successfully.

```
# ls -ld /var/www/html
# cat /webdev/index.html
# systemctl status httpd
```

```
[ec2-user@ip-172-31-42-176 ~]$ ls -ld /var/www/html
lrwxrwxrwx. 1 root root 7 Jun 24 10:10 /var/www/html -> /webdev
[ec2-user@ip-172-31-42-176 ~]$ cat /webdev/index.html
development
```

```
[ec2-user@ip-172-31-42-176 ~]$ sudo systemctl status httpd
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
   Drop-In: /usr/lib/systemd/system/httpd.service.d
            └─php-fpm.conf
   Active: active (running) since Wed 2026-06-24 10:10:27 UTC; 1min 50s ago
     Docs: man:httpd.service(8)
  Main PID: 43031 (httpd)
    Status: "Total requests: 0; Idle/Busy workers 100/0; Requests/sec: 0; Bytes served/sec: 0 B/sec"
    Tasks: 177 (limit: 1059)
   Memory: 18.3M
      CPU: 172ms
   CGroup: /system.slice/httpd.service
           └─43031 /usr/sbin/httpd -DFOREGROUND
             └─43033 /usr/sbin/httpd -DFOREGROUND
               └─43034 /usr/sbin/httpd -DFOREGROUND
                 └─43035 /usr/sbin/httpd -DFOREGROUND
                   └─43066 /usr/sbin/httpd -DFOREGROUND

Jun 24 10:10:27 ip-172-31-42-176.ap-south-1.compute.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
Jun 24 10:10:27 ip-172-31-42-176.ap-south-1.compute.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
Jun 24 10:10:27 ip-172-31-42-176.ap-south-1.compute.internal httpd[43031]: Server configured, listening on: port 80
[ec2-user@ip-172-31-42-176 ~]$ curl http://localhost
development
[ec2-user@ip-172-31-42-176 ~]$
```

## Task 7: Git Integration

### Requirements

1. Configure the local Git repository.
2. Add the required Ansible files and role to the repository.
3. Commit the project files with an appropriate commit message.
4. Push the committed changes to the remote Git repository.
5. Verify the uploaded files and Git commit history in the repository.

## Step 1: Create and Configure the GitHub Repository

Initialize the local GitHub repository, add the required project files, and commit the changes with an appropriate commit message.

```
# git commit -m "Initial Ansible Automation Project"
```

Creates a commit and saves the project changes to the local Git repository

```
[devops@ip-172-31-43-214 ansible]$ git commit -m "Initial Ansible Automation Project"
[master (root-commit) f68515e] Initial Ansible Automation Project
 8 files changed, 130 insertions(+)
 create mode 100644 ansible.cfg
 create mode 100644 custom.yml
 create mode 100644 inventory
 create mode 100644 issue.yml
 create mode 100644 myrole.yml
 create mode 100644 packages.yml
 create mode 100644 roles/myrole/tasks/main.yml
 create mode 100644 roles/myrole/templates/index.j2
[devops@ip-172-31-43-214 ansible]$
```

## Step 2: Push the Project to GitHub

Configure the remote repository and push the committed project files to the GitHub repository.

```
# git push -u origin main
```

Pushes the committed project files from the local repository to the GitHub repository

```
[devops@ip-172-31-43-214 ansible]$ git branch -M main
[devops@ip-172-31-43-214 ansible]$ git push -u origin main
Username for 'https://github.com': nandusivadas98@gmail.com
Password for 'https://nandusivadas98%40gmail.com@github.com':
Enumerating objects: 14, done.
Counting objects: 100% (14/14), done.
Delta compression using up to 2 threads
Compressing objects: 100% (11/11), done.
Writing objects: 100% (14/14), 1.61 KiB | 1.61 MiB/s, done.
Total 14 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote: This repository moved. Please use the new location:
remote:   https://github.com/nandusivadas/Ansible-Automation-Project.git
To https://github.com/nandusivadas/ansible-automation-project.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
[devops@ip-172-31-43-214 ansible]$
```

## Step 3: Verify the Repository

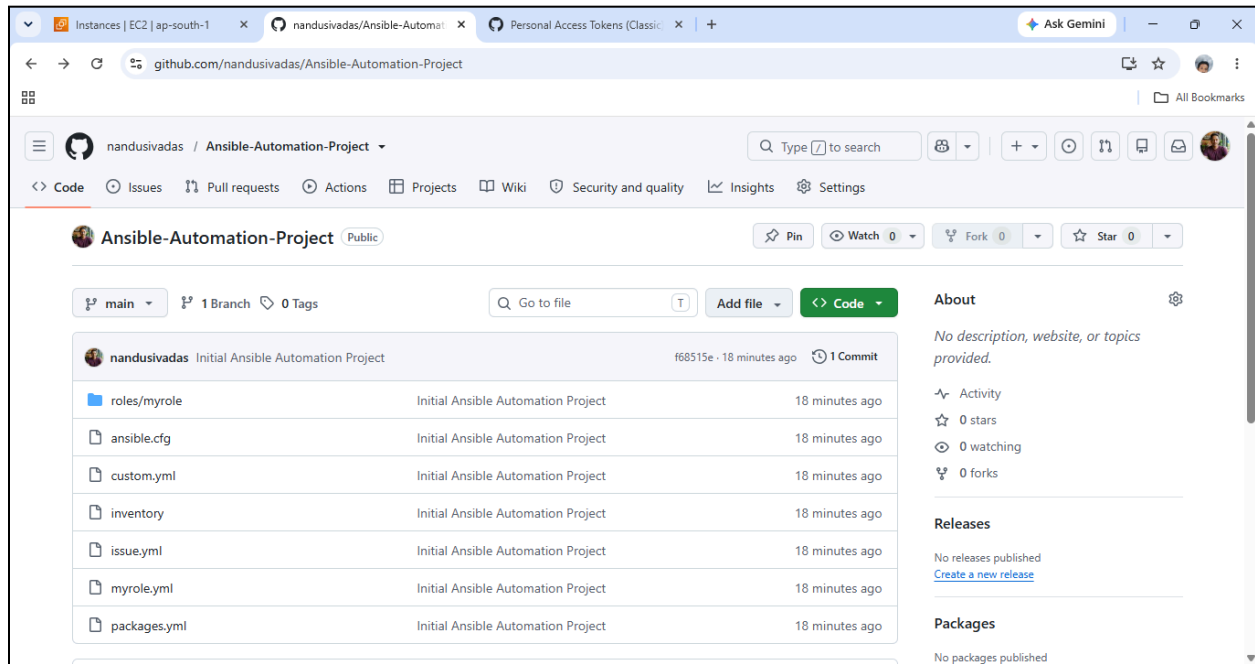
Verify the Git commit history and confirm that all project files have been uploaded successfully to the GitHub repository.

```
# git log --oneline
```

Displays the Git commit history in a single-line format to verify the commit.

```
[devops@ip-172-31-43-214 ansible]$ git log --oneline
f68515e (HEAD -> main, origin/main) Initial Ansible Automation Project
[devops@ip-172-31-43-214 ansible]$
```

GitHub repository page showing the uploaded files



## Mission 1 Completion

Mission 1 was successfully completed by implementing all the required tasks and verifying the expected outputs. The Ansible environment was configured successfully, connectivity between the control node and client nodes was established, and all project requirements were achieved.

# MISSION 2 – Infrastructure Recreation Using Git

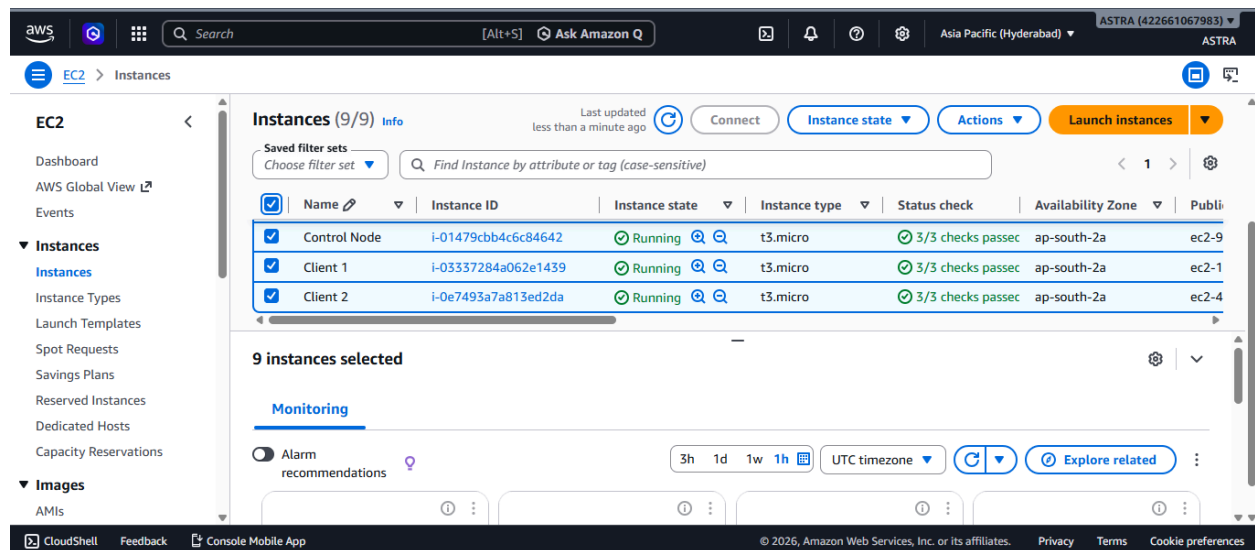
## Phase 1: Create a New Environment

### Requirements

1. Launch a new AWS environment in the **Hyderabad Region**.
2. Create one control node and two client nodes.
3. Assign the required instance names:
  - `hyderabad-control1`
  - `hyderabad-client1`
  - `hyderabad-client2`
4. Create a user named `c1one` on all machines.
5. Create the Ansible project directory at `/home/c1one/ansible`

### Step 1: Launch the AWS Environment

Launch one control node and two client nodes in the **AWS Hyderabad Region** using the required naming convention.



### Step 2: Create the c1one User

Create the `c1one` user on the **control node** and **both client nodes**, then add the user to the `wheel` group for administrative privileges. Verify the user creation using the `id c1one` command.

```
ec2-user@ip-172-31-6-129:~  
login as: ec2-user  
Authenticating with public key "imported-openssh-key"  
#  
_##### Amazon Linux 2023  
~~~\#####\br/>~~~\###|br/>~~~\#/ https://aws.amazon.com/linux/amazon-linux-2023  
~~~V~'~>  
~~~  
~~~  
~~~  
~~~  
[ec2-user@ip-172-31-6-129 ~]$ sudo useradd clone  
[ec2-user@ip-172-31-6-129 ~]$ sudo usermod -aG wheel clone  
[ec2-user@ip-172-31-6-129 ~]$ id clone  
uid=1001(clone) gid=1001(clone) groups=1001(clone),10(wheel)  
[ec2-user@ip-172-31-6-129 ~]$
```

### Step 3: Configure Passwordless SSH Authentication

Generate an SSH key pair on the control node and configure passwordless SSH access to both client nodes.

In Control Node:

# ssh-keygen

```
[ec2-user@ip-172-31-6-129 ~]$ sudo su - clone  
[clone@ip-172-31-6-129 ~]$ ssh-keygen  
Generating public/private rsa key pair.  
Enter file in which to save the key (/home/clone/.ssh/id_rsa):  
Created directory '/home/clone/.ssh'.  
Enter passphrase (empty for no passphrase):  
Enter same passphrase again:  
Your identification has been saved in /home/clone/.ssh/id_rsa  
Your public key has been saved in /home/clone/.ssh/id_rsa.pub  
The key fingerprint is:  
SHA256:kQEGr9onht9iYRwMSYdRYbops89GEuLQo5QNwBGg/7M clone@ip-172-31-6-129.ap-south-2.compute.internal  
The key's randomart image is:  
+---[RSA 3072]-----+  
|*+*.o...|  
|o+ B o|  
|. +B o o|  
|++*+|  
|B+o S|  
|oO=|  
|+++.|  
|++ o|  
| o+ E|  
+---[SHA256]-----+  
[clone@ip-172-31-6-129 ~]$ cat ~/.ssh/id_rsa.pub  
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAgQDn2c3r3WOGYN00stZs+pdCeD/ZvGzCb7Gks+tiLWXA5mXMG3qzNdr83LkWPz2/6YWXVWioV1NJuwA8UVhjtjfy5n5DS4vmoh  
UE6EWqulq9Xlbemj2GBCIeipylBOGA2cybuzydWPYsiyGKjgjs21Rxtn2d37N+bVhtVQN+i2NV9WuK852hzJZWms0zIlbrqTeXHIWuCTfsMO1Uqs90LQaTJLjxl0Ny8+tLGiw7  
mXPzKioYz+r27h8LShtnUzt3rdzIc+EkFa6pWdDihqAMZ0VqWMDbO8BFvJAs9bpsIXOygTQrverTP9sqVMzQb0p1T19eYMBKDJp8cVCKUj9lYa3qC8nqAyo/HYaYdpA7qmSN14  
p9EYuk4ktoSzJ7iN7pqWVUQrBBglhf1E/yeEikP3e2VZGuYxU6Gt21+jBEii662HxPwufORGGXTmrhHHh72F5BhALg7h8KRiABEWifMRhdYmmtulVjzYr5ov1PQ/rGbay6X1  
VxXNwGqgBVEJONs= clone@ip-172-31-6-129.ap-south-2.compute.internal  
[clone@ip-172-31-6-129 ~]$
```

### Step 4: Configure SSH Authentication on Client Nodes

Create the `.ssh` directory and configure the `authorized_keys` file on the client nodes to enable passwordless SSH authentication.

```
[ec2-user@ip-172-31-15-188 ~]$ sudo su - clone
[clone@ip-172-31-15-188 ~]$ mkdir -p ~/.ssh
[clone@ip-172-31-15-188 ~]$ chmod 700 ~/.ssh
[clone@ip-172-31-15-188 ~]$ vim ~/.ssh/authorized_keys
[clone@ip-172-31-15-188 ~]$ chmod 600 ~/.ssh/authorized_keys
[clone@ip-172-31-15-188 ~]$
```

## Successful SSH login from Control Node to a Client node

[illegible]

## Step 5: Create the Ansible Project Directory

Create the project directory `/home/clone/ansible` on the control node.

```
[clone@ip-172-31-6-129 ~]$ mkdir ~/ansible
[clone@ip-172-31-6-129 ~]$ cd ~/ansible
```

## Phase 2: Clone Existing Automation Project

## Requirements

1. Clone the Git repository into the Ansible project directory.
2. Update the inventory file to match the new environment.
3. Verify SSH connectivity between the control node and client nodes.
4. Execute all Ansible playbooks using Ansible Navigator.
5. Verify the successful execution of all playbooks:
  - `packages.yml`
  - `myrole.yml`
  - `issue.yml`

- o custom.yml

## Step 1: Configure Docker Environment

Installed Docker on the control node, enabled and started the Docker service, and verified that it was running successfully to support the Ansible Execution Environment.

```
[clone@ip-172-31-6-129 ansible]$ sudo systemctl enable docker
Created symlink /etc/systemd/system/multi-user.target.wants/docker.service → /usr/lib/systemd/system/docker.service.
[clone@ip-172-31-6-129 ansible]$ sudo systemctl start docker
[clone@ip-172-31-6-129 ansible]$ sudo systemctl status docker
● docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: disabled)
   Active: active (running) since Thu 2026-06-25 07:42:00 UTC; 14s ago
 TriggeredBy: ● docker.socket
     Docs: https://docs.docker.com
    Process: 28375 ExecStartPre=/bin/mkdir -p /run/docker (code=exited, status=0/SUCCESS)
    Process: 28376 ExecStartPre=/usr/libexec/docker/docker-setup-runtimes.sh (code=exited, status=0/SUCCESS)
   Main PID: 28377 (dockerd)
      Tasks: 9
     Memory: 30.0M
        CPU: 336ms
     CGroup: /system.slice/docker.service
             └─28377 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock --default-ulimit nofile=32768:65536

Jun 25 07:41:59 ip-172-31-6-129.ap-south-2.compute.internal systemd[1]: Starting docker.service - Docker Application Container Engine.
Jun 25 07:41:59 ip-172-31-6-129.ap-south-2.compute.internal dockerd[28377]: time="2026-06-25T07:41:59.687076404Z" level=info msg="Starting daemon"
Jun 25 07:41:59 ip-172-31-6-129.ap-south-2.compute.internal dockerd[28377]: time="2026-06-25T07:41:59.757358535Z" level=info msg="Loading containers"
Jun 25 07:42:00 ip-172-31-6-129.ap-south-2.compute.internal dockerd[28377]: time="2026-06-25T07:42:00.296831210Z" level=info msg="Loading containers: done."
Jun 25 07:42:00 ip-172-31-6-129.ap-south-2.compute.internal dockerd[28377]: time="2026-06-25T07:42:00.330468670Z" level=info msg="Daemon has started"
Jun 25 07:42:00 ip-172-31-6-129.ap-south-2.compute.internal dockerd[28377]: time="2026-06-25T07:42:00.359211057Z" level=info msg="API endpoint is now available"
Jun 25 07:42:00 ip-172-31-6-129.ap-south-2.compute.internal systemd[1]: Started docker.service - Docker Application Container Engine.
lines 1-22/22 (END)
```

## Step 2: Install and Verify Ansible

Installed Ansible and Ansible Navigator on the control node and verified the installation

```
[clone@ip-172-31-6-129 ansible]$ ansible --version
ansible [core 2.15.13]
  config file = None
  configured module search path = ['/home/clone/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']
  ansible python module location = /home/clone/.local/lib/python3.9/site-packages/ansible
  ansible collection location = /home/clone/.ansible/collections:/usr/share/ansible/collections
  executable location = /home/clone/.local/bin/ansible
  python version = 3.9.25 (main, May 25 2026, 00:00:00) [GCC 11.5.0 20240719 (Red Hat 11.5.0-5)] (/usr/bin/python3)
  jinja version = 3.1.6
  libyaml = True
[clone@ip-172-31-6-129 ansible]$ ansible-navigator --version
ansible-navigator 24.2.0
[clone@ip-172-31-6-129 ansible]$
```

## Step 3: Clone the Automation Project

Cloned the Git repository into the `/home/clone/ansible` project directory.

```
[clone@ip-172-31-6-129 ansible]$ git clone https://github.com/nandusivadas/Ansible-Automation-Project.git .
Cloning into '.'...
remote: Enumerating objects: 14, done.
remote: Counting objects: 100% (14/14), done.
remote: Compressing objects: 100% (11/11), done.
remote: Total 14 (delta 0), reused 14 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (14/14), done.
[clone@ip-172-31-6-129 ansible]$ ls
ansible.cfg  custom.yml  inventory  issue.yml  myrole.yml  packages.yml  roles
[clone@ip-172-31-6-129 ansible]$
```

## Step 4: Configure the Inventory File

Updated the Ansible inventory file with the hostnames and IP addresses of the client nodes.

```
[clone@ip-172-31-6-129 ansible]$ vim inventory
[clone@ip-172-31-6-129 ansible]$ cat inventory
[dev]
hyderabad-client1 ansible_host=172.31.15.188

[test]
hyderabad-client2 ansible_host=172.31.4.20

[all:vars]
ansible_user=clone
[clone@ip-172-31-6-129 ansible]$
```

## Step 5: Verify SSH Connectivity

Verified passwordless SSH connectivity between the control node and the client nodes using the Ansible ping playbook.

```
clone@ip-172-31-6-129:~/ansible
[clone@ip-172-31-6-129 ansible]$ ansible-navigator run ping.yml --mode stdout
-----
Execution environment image and pull policy overview
-----
Execution environment image name:  ghcr.io/ansible/creator-ee:v0.22.0
Execution environment image tag:   v0.22.0
Execution environment pull arguments: None
Execution environment pull policy:  tag
Execution environment pull needed:  True
-----
Updating the execution environment
-----
Running the command: docker pull ghcr.io/ansible/creator-ee:v0.22.0
v0.22.0: Pulling from ansible/creator-ee
ca6ff2c3ecc4: Pull complete
2941e6f1d2bc: Pull complete
32db66e6bbce: Pull complete
92e1f4a92098: Pull complete
600b306c9749: Pull complete
2764fac24a7d: Pull complete
d419ad167630: Pull complete
7b03b9e3c798: Pull complete
05a232cb5cea: Pull complete
be915b595e0a: Pull complete
25380a2f785a: Pull complete
1a57afb20498: Pull complete
09f2372dbd92: Pull complete
Digest: sha256:2edd3559b9fa2f3a386d078bff531874a1d30455f8171d027cf3a9c7f50118d4
Status: Downloaded newer image for ghcr.io/ansible/creator-ee:v0.22.0
ghcr.io/ansible/creator-ee:v0.22.0

PLAY [all] *****

TASK [Verify SSH Connectivity] *****
[WARNING]: Platform linux on host hyderabad-client2 is using the discovered
Python interpreter at /usr/bin/python3.9, but future installation of another
Python interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-
```

```
core/2.16/reference_appendices/interpreter_discovery.html for more information.
ok: [hyderabad-client2]
[WARNING]: Platform linux on host hyderabad-client1 is using the discovered
Python interpreter at /usr/bin/python3.9, but future installation of another
Python interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-
core/2.16/reference_appendices/interpreter_discovery.html for more information.
ok: [hyderabad-client1]

PLAY RECAP *****
hyderabad-client1      : ok=1    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```



## Step 6: Execute the Required Playbooks

Executed the required Ansible playbooks to configure the client nodes and install the necessary packages.

```
[clone@ip-172-31-6-129 ansible]$ ansible-navigator run packages.yml --mode stdout

PLAY [dev] *****

TASK [Gathering Facts] *****
[WARNING]: Platform linux on host hyderabad-client1 is using the discovered
Python interpreter at /usr/bin/python3.9, but future installation of another
Python interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-core/2.16/reference\_appendices/interpreter\_discovery.html for more information.
ok: [hyderabad-client1]

TASK [Update all packages] *****
ok: [hyderabad-client1]

TASK [Install mariadb] *****
changed: [hyderabad-client1]

TASK [Install php] *****
changed: [hyderabad-client1]

TASK [Install Development Tools] *****
changed: [hyderabad-client1]

PLAY [test] *****

TASK [Gathering Facts] *****
[WARNING]: Platform linux on host hyderabad-client2 is using the discovered
Python interpreter at /usr/bin/python3.9, but future installation of another
Python interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-core/2.16/reference\_appendices/interpreter\_discovery.html for more information.
ok: [hyderabad-client2]

TASK [Install mariadb] *****
changed: [hyderabad-client2]

TASK [Install php] *****
changed: [hyderabad-client2]

PLAY RECAP *****
hyderabad-client1      : ok=5    changed=3    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
hyderabad-client2      : ok=3    changed=2    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
[clone@ip-172-31-6-129 ansible]$
```

## Step 7: Deploy the Apache Web Server

Executed the Ansible role to install and configure the Apache web server on the target node.

```
[clone@ip-172-31-6-129 ansible]$ ansible-navigator run myrole.yml --mode stdout

PLAY [test] *****

TASK [Gathering Facts] *****
[WARNING]: Platform linux on host hyderabad-client2 is using the discovered
Python interpreter at /usr/bin/python3.9, but future installation of another
Python interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-core/2.16/reference\_appendices/interpreter\_discovery.html for more information.
ok: [hyderabad-client2]

TASK [myrole : Install Apache] *****
ok: [hyderabad-client2]

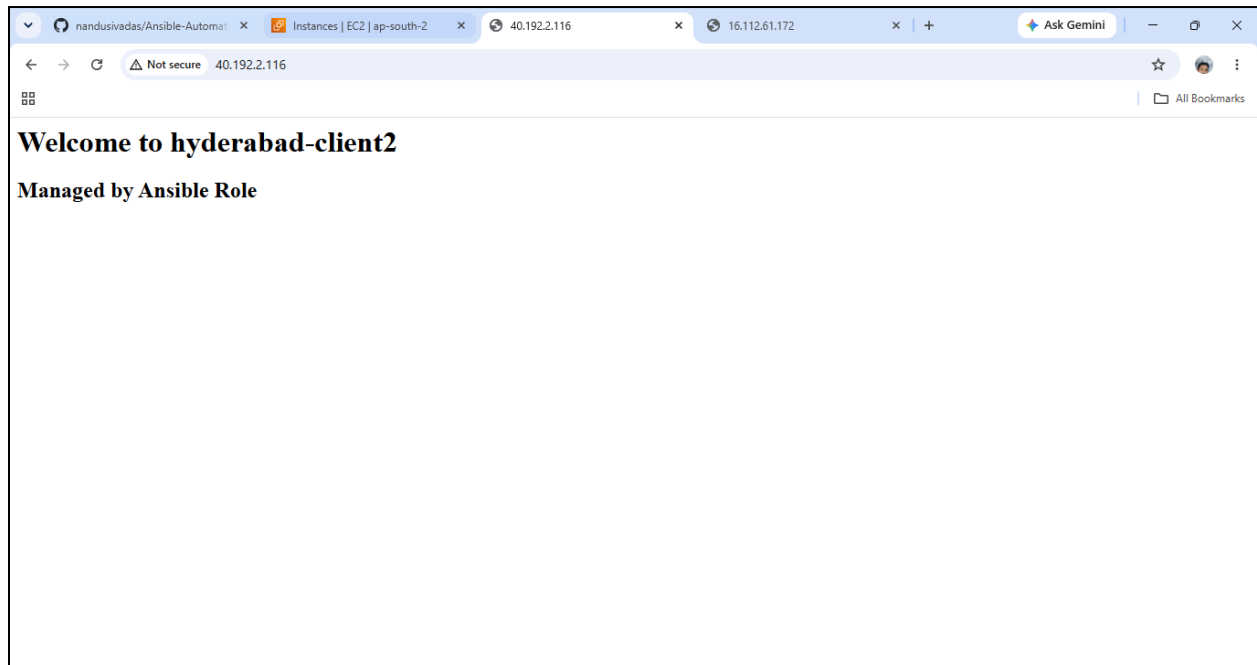
TASK [myrole : Start Apache] *****
changed: [hyderabad-client2]

TASK [myrole : Deploy index page] *****
changed: [hyderabad-client2]

PLAY RECAP *****
hyderabad-client2      : ok=4    changed=2    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
[clone@ip-172-31-6-129 ansible]$
```

## Step 8: Verify the Web Server Output

Verified the successful deployment by accessing the web application through a web browser.



## Mission 2 Completion

Mission 2 was successfully completed by cloning the automation project, configuring the inventory, verifying SSH connectivity, and executing all required Ansible playbooks using Ansible Navigator. The Apache web server was deployed successfully, and the web application outputs were verified on the client nodes, confirming that all project requirements were achieved.

## Project Outcome

The project successfully automated the configuration and management of multiple Linux servers using Ansible. All required playbooks and roles were executed successfully, and the expected outputs were verified on the target systems.

# Conclusion

The Ansible Automation Project was completed successfully by setting up the control node, configuring passwordless SSH communication, creating and executing playbooks, and deploying web server configurations using Ansible roles. The project demonstrated how automation simplifies system administration by reducing manual effort and ensuring consistent configuration across multiple servers.

## Skills Gained

- AWS EC2 Instance Management
- Linux User Administration
- SSH Key-Based Authentication
- Git and GitHub
- Docker
- Ansible
- Ansible Navigator
- Ansible Playbooks
- Ansible Roles
- Apache Web Server Deployment

| Verification Summary        |        |           |
|-----------------------------|--------|-----------|
| Task                        | Status |           |
| ✓ AWS Environment Created   | ✓      | Completed |
| ✓ Control Node Configured   | ✓      | Completed |
| ✓ Client Nodes Configured   | ✓      | Completed |
| ✓ SSH Connectivity Verified | ✓      | Completed |
| ✓ Git Repository Cloned     | ✓      | Completed |
| ✓ Playbooks Executed        | ✓      | Completed |
| ✓ Apache Deployment         | ✓      | Completed |
| ✓ Web Server Verified       | ✓      | Completed |

# GitHub Repository

Repository URL:

<https://github.com/nandusivadas/Ansible-Automation-Project>

The complete source code, configuration files, documentation, and project resources are available in the GitHub repository.

## References

- AWS Documentation
  - Ansible Documentation
  - Docker Documentation
  - Git Documentation
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